

Mathematics I

CSIT — 1st Semester — 2081 — Answer Sheet
Subject Code: MTH117

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. (a) Sketch the graph of $f(x) = x^2$. Find its domain and range. (b) Evaluate: $\lim_{x \rightarrow 1} \sin^{-1} \left(\frac{1 - \sqrt{x}}{1 - x} \right)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. (a) Where the function $f(x) = |x|$ is differentiable? Discuss. (b) A farmer has 1200 m. of fencing and wants to fence off a rectangular field that borders a straight river. He needs to fence along the river. What are the dimensions of the field that has the largest area?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. (a) Find the solution of the initial value problem $(x^2 y' + xy = 1)$, $(y(1) = 2)$, $(x > 0)$. (b) Find the area enclosed by the line $(y = x - 1)$ and the parabola $(y^2 = 2x + 6)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Evaluate: $\int_0^{\sqrt{3}} \sqrt{1+x^2} \, dx$, $\int x^3 \, dx$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find the Maclaurin series expansion of $f(x) = \sin x$ for all x .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Find the unit normal and binormal vectors for the circular helix $\mathbf{r}(t) = \cos t \, \mathbf{i} + \sin t \, \mathbf{j} + t \, \mathbf{k}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. If $f(x, y) = \frac{xy}{x^2 + y^2}$, does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist? Justify.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Determine whether the sequence $(a_n = (-1)^n)$ is convergent or divergent.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. The position vector of an object moving in a plane is given by $\mathbf{r}(t) = t^3 \mathbf{i} + t^2 \mathbf{j}$. Find its velocity, speed, and acceleration when $(t = 1)$ and illustrate geometrically.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Show that every member of the family of function $y = \frac{1 + ce^t}{1 - ce^t}$ is a solution of the differential equation $(y' = \frac{1}{2}(y^2 - 1))$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. If $f(x, y) = 2x^3 - x^2y^3 - y^4$, find $f_x(1, -2)$, $f_y(1, -1)$ and $f_{yx}(1, -1)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Use cylindrical shells to find the volume of the solid obtained by rotating about the x-axis the region under the curve $(y = \sqrt{x})$ for 0 to 1.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.